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MAGNETIC STRATIGRAPHY AND MAGNETIC-PETROLOGIC PROPERTIES OF PRECAMBRIAN GARDAR LAVAS, SOUTH GREENLAND

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The Gardar lavas (<1600 m.y., >1310 m.y.) within the Eriksfjord Formation of South Greenland were a prelude to major hypabyssal magmatism within the Gardar alkaline igneous province. This paper reports a palaeomagnetic study of the lower two-thirds of the lava stratigraphy comprising three groups of flows separated by sediments. The lower group (palaeomagnetic pole at 239°E, 7°N, 7 sites, $d\Psi = 7.6°$, $d\chi = 14.1°$) includes one reversal of magnetisation and two intermediate directions, and the upper group (pole at 222°E, 16°N, 24 sites, $d\Psi = 4.8°$, $d\chi = 9.0°$) includes four reversals and two intermediate directions. The middle group of 13 lavas yields anomalous directions with respect to the other groups and appears to have been erupted during a polarity excursion of uncertain significance. Opaque petrographic analysis reveals all stages of deuteric oxidation (800–500°C) plus lower-temperature hydrothermal alteration to epidote grade. In addition to high reversal frequency, the succession illustrates asymmetry of the dipole source, and there is a suggestion that lavas with easterly directed magnetisations have higher deuteric oxidation states than those with westerly directed magnetisations.

Collective palaeomagnetic results from the Gardar Province are linked to a number of Rb–Sr isochrons, and comparison with contemporaneous North American data confirms the 500-fathom continental reconstruction for Proterozoic times. The poles define a movement of the Laurentian craton from low palaeolatitudes (1400–1220 m.y.) to high palaeolatitudes (1168–1070 m.y.) with a quasi-static interval at 1280–1220 m.y. coinciding with the Mackenzie igneous episode. The youngest Gardar data correlate with the oldest Keweenawan palaeomagnetic poles and restrict deposition of this group to the interval ca. 1160–1040 m.y.

1. Introduction

The Proterozoic Gardar igneous cycle of South Greenland [1] commenced with eruption of basaltic lavas and deposition of continental sandstones (Eriksfjord Formation), nearly contemporary with, or followed by, intrusion of major alkaline igneous bodies, syenite and gabbro giant dykes and a whole range of acid to basic minor intrusions. The lavas within the Eriksfjord Formation have an aggregate thickness of over 1500 m [2]; not only are they well exposed, but intervening clastic horizons show that the extrusion of the earlier phases spanned a considerable period of time. They provide one of the limited number of opportunities to study the sequence of magnetic field behaviour in a Precambrian lava succession. This paper reports the palaeomagnetic and opaque petrologic properties of a stratigraphically controlled sample through approximately the lower two-thirds of the succession (Fig. 1).

2. Geology and age

The Eriksfjord Formation [3] is largely preserved within a ENE-trending fault–bounded block and subdivides into three members (Mâjût, Naujarssuit and Nunarsarnaq Sandstones) separated by lavas. Following a basal conglomerate, the two lower members are bedded red sandstones which appear to have been deposited mostly, if not entirely, in shallow water; the upper member however, has large-scale cross-bedding units and dune relicts and appears to be a subaerial deposit. Of the lava divisions [2,4], the lowest (Mussartût member) comprises a number of aphyric volcanic members separated by red sandstones and

conglomerates; the former were originally regarded as sills [2] but have subsequently been reinterpreted as lavas [5]. The next (Ulukasik) member comprises 250 m of rapidly extruded aphyric and feldspar-phyric units with oxidised, ropy and scoriaceous surfaces and cavities filled by wind-blown sand. The general absence of pyroclastics and the pahoehoe nature of these flows suggest that they derive from fissures or low shield volcanoes. The upper (Ilímaussaq) member is the thickest (ca. 1100 m) and the most varied, commencing with "star" basalts (with distinctive radiating groups of white plagioclase phenocrysts) followed by aphyric basalts, a "platey" (feldspar-phyric) subdivision, and merging into trachytic lavas. The sample for this study includes all members of the lower two divisions and a section to the top of the "platey" basalt division in the Ilímaussaq member; there are approximately 700 m of basaltic and intermediate flows above this.

The lava sequence is now preserved between Tugtutôq to the south and the inland ice to the north [6]; it was probably once more widespread since xenoliths from the succession occur in Gardar diatremes in and around the Tugtutôq-Narssaq area and within the central Tugtutôq alkaline complex to the south-west [6,7].

The underlying granite/granite-gneiss basement

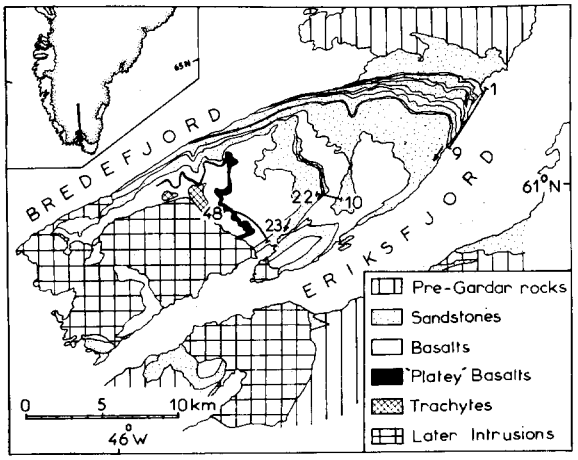


Fig. 1. Sampling sites (indicated by numbers) and geology of the Eriksfjord Formation and Gardar lavas, South Greenland, slightly simplified after 1 : 100,000 maps of Greenland 60V 2 Nord (Julianehaab) and 61V 3 Syd (Narssarssuaq), and based on geological mapping of Stewart [2].

belongs to the Ketilidian mobile belt with a final cooling age of ca. 1600 m.y. [8] and outliers are intruded by the (Gardar) Motzfeldt centre of the Igaliko alkaline complex for which Blaxland et al. [9] determined a Rb–Sr whole-rock isochron of 1310 ± 31 m.y. The Eriksfjord succession was therefore deposited sometime between 1600 and 1310 m.y., but its spatial relationship and chemical affinities with the Gardar Province (1330–1160 m.y.) suggest that an age near the younger of these two limits it most likely.

3. Palaeomagnetic results

Up to six samples were collected from each lava flow or flow unit using a two-stroke portable drill [10]; cores were oriented while still attached to the outcrop by sights on the sun except for sites 23–32 where a magnetic compass had to be employed in conjunction with topographic sightings. The locations within the lava succession were recorded by altimeter. In the laboratory the cores were sliced into 2.5-cm cylinders and measured before and after progressive alternating field (a.f.) demagnetisation using a parastatic magnetometer; the step-wise a.f. cleaning was normally undertaken in steps of 50 oersteds (Oe) up to maximum peak fields between 500 and 1400 Oe. Curie points were determined for a small fragment (approximately 1 mm diameter) from each lava using a modified Chevallier-type balance and heating in air with a strong permanent magnet to saturate the specimen. Finally, a polished section was made from a slice of the base of one core from each site and utilised for petrographic study under reflected light with a Reichert Zetopan-Pol microscope at a magnification of X1200 under oil immersion.

For the palaeomagnetic analysis, samples from the lower group were corrected for a tectonic dip of 9° in a direction 180°E, and the samples from the middle and upper groups were corrected for a dip of 10° in a direction 220°E determined from the interleaved sediments.

As a measure of the magnetic hardness, the range of the parameter M_{200}/M_0 (ratio of moment after treatment in 200 Oe to primary moment) is included in Table 1; it illustrates the wide range of magnetic coercivities frequently present in one site. In the lower

(Mussartût) group most samples show exponential fall in moments with progressive treatment, but magnetic stability is high for all ranges of coercivity, although samples with M_{200}/M_0 values less than 0.4 tend to acquire unstable components as demagnetisation proceeds above 400–500 Oe. Samples from the middle (Ulukasik) group all exhibit very high coercivities with little or no change in direction or moment, and samples from the upper (Ilímaussaq) group often exhibit a wide range of coercivities in samples from the same site (Fig. 2); some samples show constant or slight increase in moment while others show systematic exponential fall with progressive treatment. Most of these samples are, however, highly stable and yield similar or identical directions of magnetisation (Fig. 2) There are a few instances (e.g. site 42, Fig. 2) where samples of high coercivity yield slightly different directions from samples with low coercivities; in these cases the former samples being hardly affected by a.f. demagnetisation may represent chemical remanent magnetisations (CRM's) aquired slightly later than primary thermo-remanent magnetisations (TRM's).

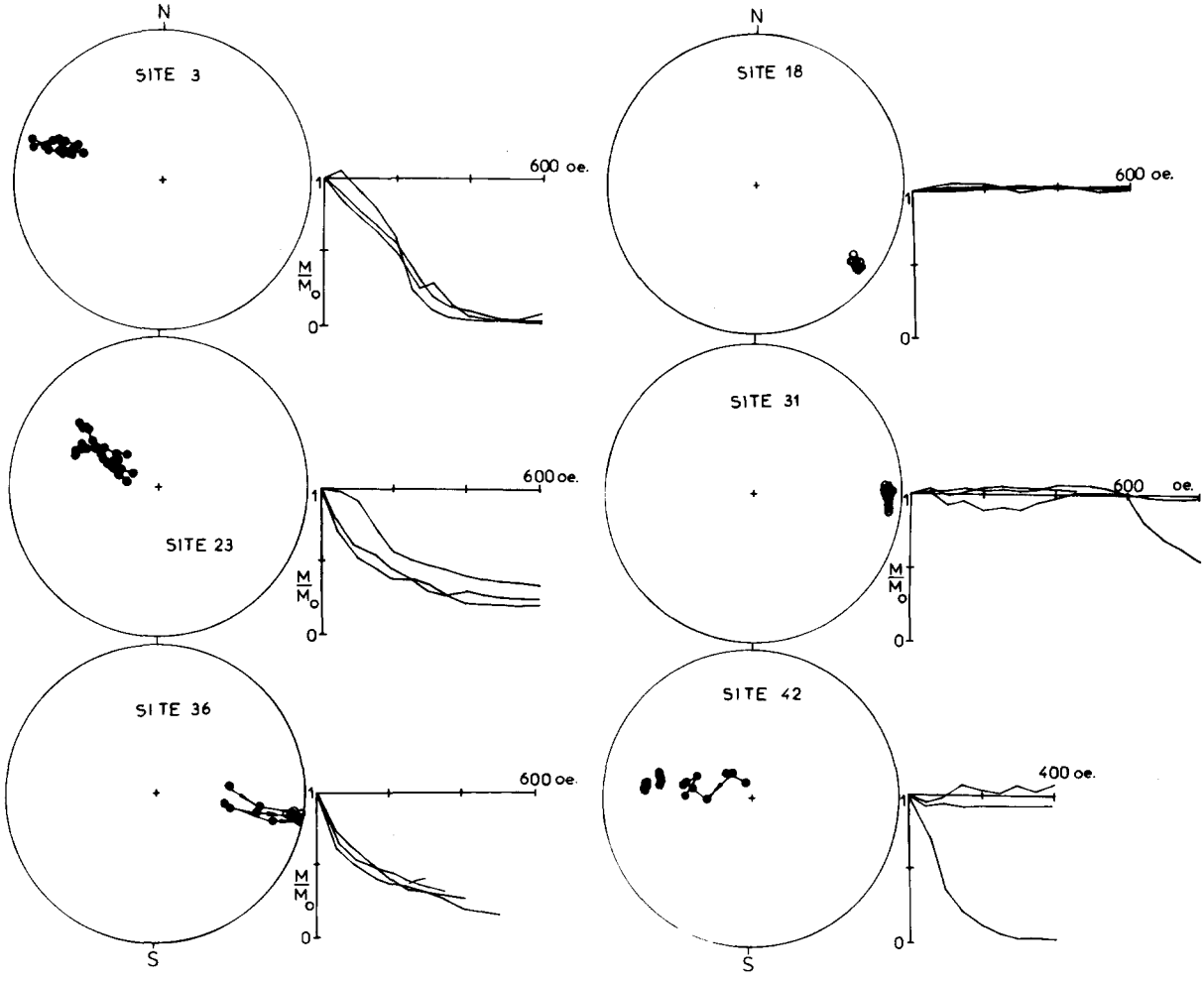


Fig. 2. Behaviour of directions of magnetisation and magnetic moments (expressed as a fraction of original moment) with progressive a.f. treatment for three samples from six lavas. Stereographic projections, open symbols are upper and closed symbols are lower hemisphere plots.

TABLE 1

Palaeomagnetic results from Precambrian Gardar lavas in the Eriksfjord Formation, South Greenland

Site No.	N	D	I	R	K	α_{95}	M_{200}/M_0	Curie point(s) ±10°C	Thermo-magnetic curve type
(1) Lower Group									
1*	5	88.6	−38.1	4.97	144	6.4	0.75–0.32	550	D
2	5	76.5	−20.1	4.89	37	12.8	0.54–0.45	530	A
3	5	283.8	15.7	4.97	116	7.1	1.04–0.60	555	B
4 ‡	5	199.5	82.9	4.61	10	24.9	1.00–0.66	535, 630	C
5	5	237.6	24.2	4.94	63	9.8	0.50–0.38	515	D
6	5	249.2	15.4	4.86	29	14.4	0.72–0.10	510	D
7	5	273.7	38.5	4.97	122	6.9	0.24–0.10	515	D
8	4	259.0	18.3	3.97	110	8.8	0.94–0.89	550, 670	B (x)
9 ‡	5	279.4	74.1	4.92	52	10.7	0.91–0.78	570, 670	B
(2) Middle Group									
10 ‡	6	216.2	48.3	5.94	81	7.5	1.00–0.83	670	A
11	5	150.0	51.6	4.98	184	5.7	0.69–0.56	540, 640	A
12	5	147.7	45.9	4.99	1569	1.9	1.02–0.96	640, 670	D
13	4	152.2	40.0	3.99	308	5.3	1.06–0.95	530, 640	D
14	5	154.7	47.6	4.96	102	7.6	1.00–0.93	540, 640	A
15	5	150.4	47.0	4.99		3.9	1.03–0.95	535	C
16	4	285.3	−28.6	3.99	306	5.3	1.18–1.01	530, 610	C
17	5	134.6	−9.4	4.76	16	19.5	1.03–1.01	620	D
18	2	126.1	−12.7	1.99	657	9.8	1.02–0.97	540, 670	C
19 ‡	5	120.3	52.6	4.94	64	9.6	0.22–0.08	575, 640	B
20	5	124.7	−10.0	4.75	16	19.6	1.06–0.84	545, 670	B
21	5	153.4	−3.5	4.26	5	36.2	1.00–0.63	535, 640	B (x)
22	6	138.8	−27.7	5.81	27	13.3	1.00–0.67	585, 670	D
(3) Upper Group									
23	4	302.5	39.0	3.97	103	9.1	0.54–0.36	540	B (x) e
24	5	279.5	55.2	4.74	15	20.3	0.37–0.13	530	A
25	5	301.7	39.4	4.80	20	17.6	1.41–0.32	535	A
26	5	311.7	70.1	4.94	71	9.2	0.25–0.09	530	A
27	3	303.1	39.7	2.90	20	28.0	1.00–0.60	570, 620	D e
28	5	298.5	20.9	4.99	547	3.3	1.05–0.89	670	C e
29	4	81.3	−11.9	3.67	9	32.2	1.00–0.50	600	C
30 **	5	100.2	−12.3	4.88	34	13.3	1.12–0.74	575	D (x) e
31	5	89.7	−7.6	4.98	227	5.1	1.04–0.98	535	A (x)
32	5	97.7	−22.9	4.74	15	20.1	1.11–0.35	605	D
33	5	103.0	−16.1	4.92	53	10.6	1.09–1.01	620	D
34	5	103.5	−15.2	4.99	534	3.3	1.05–1.04	620	C (x) e
35	5	108.2	5.7	4.84	25	15.5	0.51–0.33	500	B
36	5	99.1	−1.2	4.97	194	5.5	0.53–0.34	515	A (x)
37	5	101.7	0.3	4.99	279	4.6	0.65–0.01	590	B (x) e
38 ‡	5	284.4	72.2	4.91	47	11.2	0.51–0.15	495	A
39	4	272.2	16.5	3.86	22	20.1	1.05–0.24	530, 620	A (x)
40	5	281.2	35.1	4.87	30	14.1	1.00–0.08	550	C
41	5	277.6	34.8	4.82	22	16.7	1.09–0.83	525	A
42	5	278.1	50.7	4.60	10	25.5	1.01–0.22	570, 630	C
43	5	250.8	46.2	4.46	7	30.2	1.02–0.67	580	D
44 ***	5	82.8	0.5	4.89	37	12.7	1.16–0.97	550, 625	D
45 ***	4	90.9	3.4	3.92	37	15.3	1.03–0.86	635	D
46 ***	5	97.6	−0.3	4.99	381	3.9	1.96–1.01	570, 620	D
47	4	283.0	45.7	3.56	7	37.9	3.82–0.45	545, 615	A
48	5	266.8	54.4	4.72	14	21.0	0.40–0.05	515	A

TABLE 1 (continued)

Site No.	N	D	I	R	K	α_{95}			
Group mean palaeomagnetic results							*Palaeomagnetic pole*		
(1) Lower group	7	260.0	25.0	6.73	22	13.1	239.0°E	7.1°N	($d\Psi$ = 7.6 $d\chi$ = 14.1)
(2) Middle group									
(A)	5	151.0	46.4	4.99	317	4.3	339.7°E	1.8°N	($d\Psi$ = 3.6 $d\chi$ = 5.5)
(B)	5	135.5	−12.8	4.87	31	13.9	5.3°E	−26.3°N	($d\Psi$ = 7.2 $d\chi$ = 14.2)
(3) Upper group	24	279.5	24.0	22.3	13	8.4	222.1°E	15.6°N	($d\Psi$ = 4.8 $d\chi$ = 9.0)

The resultant vector (R) of N samples or sites is defined by a declination (D) and inclination (I); the precision parameter $k = (N - 1)/(N - R)$ and α_{95} is the semi-angle of the cone of 95% confidence about the mean direction. The palaeomagnetic poles are calculated from the group mean directions and $d\Psi$ and $d\chi$ are semi-axes of the oval of confidence about this pole at the 95% probability level in the colatitude direction and perpendicular to it respectively. Thermomagnetic curves are defined as: A = heating and cooling curves are similar; B = cooling curve below heating curve; C = cooling curve above heating curve; D = very large increase in saturation magnet isation on cooling. x = Kink in thermomagnetic curve at 150–200°C; e = 300–450°C inflection point. M_{200}/M_0 is the ratio of the magnetic moment after demagnetising in 200 Oe to the moment after treatment.

* Pillow lavas.
** Ignimbrite.
*** Feldspar-phyric units.
‡ Sites excluded from group mean analyses.

It is probable that this time difference is accomodated by short intervals of secular variation because the angular differences are mostly small and antiparallel directions are not observed although the frequency of reversal was high (Fig. 5).

For the most stable samples demagnetisation could be continued over the whole range of treatment with no significant change in direction of magnetisation, and the mean directions for each site listed in Table 1 have accordingly been derived by averaging sample directions over the most stable range of a.f. treatment; this range is 100–1400 Oe for the most stable samples and 150–400 Oe for the least stable samples.

4. Opaque petrology

Proterozoic lava successions often have characteristic suites of secondary minerals produced by hydrothermal alteration [11] for which temperature rather than pressure appears to have been the main controlling factor; these minerals are seldom visible in exposed Tertiary to Recent lava successions where a subdivision can be made on the basis of zeolite assemblages [12]. Thus in the Keweenawan-Portage Lake lavas (ca. 1100 m.y.) of the Lake Superior region, Stoiber and Davidson [13] recognise three alteration zones which (with low- and high-temperature stability information summarised from data given in Ade-Hall et al. [11]) are in order of increasing depth of burial: laumontite (180°, 195–220° to 300–380°C, prehnite (300° to 450°C) and epidote 300° to 400–500°C). Although no detailed studies have yet been made of secondary minerals in the Gardar lavas, epidote is developed as a hydrothermal mineral throughout the succession sampled for this study and is particularly widespread in palaeomagnetic samples from the middle and upper groups. A minimum hydrothermal temperature of 300°C is thus implied, and Upton et al. [5] note the extensive alteration in these rocks particularly in the

middle (Mussartût) member where up to 50% chlorite has apparently replaced primary glass. Of the silicates, olivine is entirely pseudomorphed and the pyroxenes largely so, and plagioclase is sericitised. An opaque petrologic assessment has accordingly been made to establish the condition of the remanence carriers, to resolve the arguments for and against remagnetisation, and to determine whether an alternative magnetic cleaning technique should be employed.

For petrographic analysis 60 grains (minimum) were measured and classified according to the scheme developed by Wilson and Watkins [14] and Ade-Hall et al. [15], and the practical results analysed by computer program as outlined in the footnote to Table 2. All stages of deuteric oxidation during initial cooling (750–400°C) are present in this collection. Grains were assigned a class number according to the degree of sub-solidus oxidation of high-Ti titanomagnetite to low-Ti titanomagnetite and ilmenite, and progressive oxidation of ilmenite lamellae; the mean class for all grains gives the oxidation number (Table 2). The classes (described in more detail elsewhere [14,15] are: 1, unoxidised; 2, less than 50% of the grain covered by ilmenite lamellae; 3, more than 50% of the grain covered by ilmenite lamellae; 4, ilmenite lamellae pseudomorphed by titanohematite and ferri-rutile; 5, pseudomorphed ilmenite lamellae with spinel exsolution rods in the magnetite; 6, primary grain entirely replaced by an aggregate of titanohematite, pseudobrookite and ferri-rutile.

The salient petrologic properties are listed in Table 2 where it is clear that several characteristics are typical of the deuteric alteration observed in Tertiary and Recent lavas. Firstly, 58% of this collection have oxidation numbers higher than 3. Secondly, the amount of separate ilmenite is often considerable; Ade-Hall and Lawley [16] find that separate ilmenite is much more abundant in lavas than dykes, and suggest that it develops as a function of oxygen fugacities in the crystallising magma. Thirdly, there is an inverse relationship between sulphide content and deuteric oxidation: sulphides are only found in samples with oxidation numbers 2.3 or less and are most abundant at site 7 exhibiting no oxidation. Fourthly, maghemitisation (not a deuteric process) of fractured grain margins is confined to samples with low oxidation numbers and is present in the subaqueous lava (site 1); maghemite has been widely reported in submarine lavas (e.g. [17]).

The petrologic effects of regional hydrothermal alteration are less tangible, and it is difficult or impossible to isolate the effects of deuteric and hydrothermal alteration in more oxidised samples [13]; thus titanohematite and ferri-rutile can be produced by both processes. Granulation, the development of ferri-rutile granules in little-oxidised grains, first ap-

TABLE 2

Opaque petrologic properties of Precambrian Gardar lavas, South Greenland

Site No.	Polarity	Intensity of magnetisation ($\times 10^{-5}$ emu/g)	Magnetite oxidation number	Magnetite volume	Mean magnetite size (μm)	Ilmenite volume	Sulphides	Hematite volume	Other observations
(1) Lower group									
1	E	1.60–15.92	1.6 (0.7)	99.7	12.1	–	0.3	–	Ma = 2–6
2	E	4.87–10.50	2.0 (1.1)	99.7	15.6	–	0.3	–	G = 0–1
3	W	7.46–25.92	1.4 (0.5)	99.8	5.2	–	0.2	–	G = 2–3
4	I	0.03– 0.12	1.7 (0.5)	78.2	14.1	8.0	–	13.8	G = 0–1
5	W	0.58– 3.60	1.2 (0.3)	99.9	45.3	–	0.1	–	G = 0–1, Ma = 1–4
6	W	0.53– 7.54	4.3 (0.7)	76.3	5.5	–	–	23.7	
7	W	0.62– 3.04	1.0 (0.0)	98.2	19.3	–	1.8	–	G = 2–3, Ma = 0.5–5
8	W	2.18–25.61	3.2 (1.4)	27.7	7.5	5.0	–	67.3	
9	I	0.35– 1.67	2.2 (0.7)	22.8	6.1	1.6	–	75.6	G = 1–2

TABLE 2 (continued)

Site No.	Polarity	Intensity of magnetisation ($\times 10^{-5}$ emu/g)	Magnetite oxidation number	Magnetite volume	Mean magnetite size (μm)	Ilmenite volume	Sulphides	Hematite volume	Other observations
(2) Middle group									
10	–	0.06– 0.10	3.7 (0.7)	29.4	24.6	18.1	–	52.5	
11	–	0.31– 0.57	4.1 (0.7)	46.3	26.0	45.2	–	8.5	
12	–	0.45– 0.62	3.0 (0.3)	55.4	14.5	30.0	–	15.6	
13	–	0.49– 0.83	3.2 (0.8)	13.6	11.3	–	–	86.4	
14	–	0.28– 0.77	3.4 (0.5)	65.7	22.9	16.1	–	18.2	
15	–	0.26– 0.45	4.3 (0.6)	51.7	20.3	19.5	–	28.8	
16	–	0.43– 0.89	2.8 (1.7)	92.7	45.1	6.0	–	1.3	G = 1–3
17	–	0.50– 0.78	2.0 (0.9)	11.8	14.8	8.7	–	79.5	
18	–	0.42– 0.82	1.6 (0.6)	25.0	16.1	3.8	–	71.2	
19	–	2.67– 6.41	2.8 (0.3)	67.9	40.3	15.4	–	16.7	
20	–	0.16– 0.33	3.2 (0.3)	29.8	58.5	1.1	–	69.1	
21	–	0.03– 1.23	3.7 (0.7)	38.0	26.5	3.8	–	58.2	
22	–	0.01– 0.05	3.3 (0.5)	54.3	18.8	1.7	–	44.0	
(3) Upper group									
23	W	1.42– 2.45	3.9 (0.62)	95.6	18.1	4.3	–	0.1	
24	W	6.47–17.04	4.0 (0.77)	100.0	27.5	–	–	–	
25	W	0.41– 4.26	3.4 (1.03)	98.3	22.1	1.6	–	–	
26	I	7.77–12.11	4.5 (0.70)	98.8	14.2	0.6	–	0.5	
27	W	0.16– 0.43	4.5 (0.6)	97.7	25.7	0.2	–	0.1	
28	W	0.19– 0.60	4.9 (1.1)	80.5	29.8	14.8	–	4.7	
29	W	0.19– 0.74	4.8 (0.7)	99.9	39.4	–	–	0.1	
31	E	0.86– 1.11	4.4 (1.5)	98.6	27.3	1.4	–	–	
32	E	0.22– 1.41	3.4 (1.1)	95.5	8.8	–	–	4.5	
33	E	0.87– 1.12	4.9 (0.9)	91.2	35.8	1.8	–	7.0	
34	E	1.25– 2.70	3.9 (1.3)	88.1	39.0	11.4	–	0.5	
35	E	0.59– 9.76	4.7 (1.6)	99.9	54.4	–	–	0.1	
36	E	5.43– 6.18	3.6 (0.7)	99.7	42.3	0.3	–	trace	
37	E	3.53– 5.68	3.2 (0.3)	94.3	53.0	5.6	–	0.1	
38	I	10.24–15.60	2.3 (0.7)	100.0	31.3	–	trace	trace	
39	W	3.25– 9.63	2.0 (0.9)	98.9	31.6	1.1	trace	–	
40	W	0.10– 0.84	1.2 (0.2)	75.4	28.5	0.4	–	24.2	G = 1–3
41	W	0.02– 1.22	1.3 (0.5)	99.8	41.3	–	0.2	–	G = 3
42	W	0.30– 5.80	4.3 (0.7)	95.2	58.9	2.0	–	2.8	
43	W	0.06– 1.38	5.2 (0.6)	96.1	33.2	0.7	–	2.2	
44	E	0.08– 0.65	5.1 (0.5)	65.2	36.2	34.3	–	0.5	
45	E	0.02– 0.64	2.2 (0.7)	76.3	9.2	1.3	–	12.4	
46	E	0.31– 1.15	5.3 (0.5)	92.6	48.5	4.4	–	3.0	
47	W	0.02– 0.13	2.0 (0.1)	74.9	25.1	5.4	–	19.7	
48	W	0.07– 7.70	2.6 (0.8)	92.1	41.0	3.0	–	4.9	

Polarities are defined in terms of shallow westerly (W) or shallow easterly (E) or intermediate (I) directions of magnetisation. The magnetite deuteric oxidation number is defined as $M_{OX} = \Sigma_{n=1}^{6}$ ($n \times$ percentage of total magnetite represented by class n) where n is any magnetite oxidation class from 1 to 6. The figure in brackets is an expression of the range of magnetite deuteric oxidation represented in the sample and is given by $\Sigma_{n=1}^{6}$ ($M_{OX} - n$) $\times$ area of $n/\Sigma_{n=1}^{6}$ area. Volumes are given as percentages of total opaque volume (opaques range from 2–16% of the rock volume in these samples. Maghemitisation (Ma) is classed as affecting 10% (1) to 100% (10) of the samples and granulation G is classed as affecting 1/3 (1) to 3/3 (3) of sample areas.

pears in zone D in eastern Iceland [15] corresponding to temperatures of about 150°C; it is sporadically observed throughout this succession in sites with low oxidation numbers. Ade-Hall et al [11] regard hematisation of magnetite and relict ilmenite in a base of titanohematite and ferri-rutile as diagnostic of hydrothermal alteration of class 3 and 4 grains in the epidote (I) zone. Hematisation is widely developed in this collection, notably in the middle group (Table 2), although since it is very finely divided the precise amount is difficult to measure. It is apparently unrelated to deuteric oxidation because samples with both low and high deuteric oxidation states are affected to comparable degrees; it is ubiquitous where the primary magnetite grain size is smallest (Table 2). This effect is therefore attributable to regional hydrothermal alteration and its significance in the context of remagnetisation is discussed in section 7.

5. Thermomagnetic characteristics

Keweenawan lavas from the epidote zone are characterised by distinctive thermomagnetic curves [11] with a (as yet unexplained) rise in saturation magnetisation (J_s) between 100 and 200°C to produce a kink (Fig. 3), and an irreversible inflection ("e" point) between 300 and 450°C. This collection is more varied and inflection points and kinks are confined to a few lavas mostly in the upper group (Table 1).

In general all curves show a single high Curie point due to titanomagnetite (495–580°C) and sometimes a slightly higher point due to hematite (590–670°C); hematite observed in polished sections corresponds well with that detected on thermomagnetic curves (Tables 1 and 2).

The manner in which hematite contributes to the magnetic moments is clearly a subtle one; thus it might be invoked to explain the high coercivities of lavas of the middle group but cannot account for the often comparable magnetic hardness of samples in the upper group. Equally, site 19 with a much higher content of hematite than site 16 has magnetic moments dominated by magnetite, while site 16 with similar deuteric oxidation state but lower hematite content illustrates a much greater magnetic hardness. Evidently the contribution of hematite to the magnetic

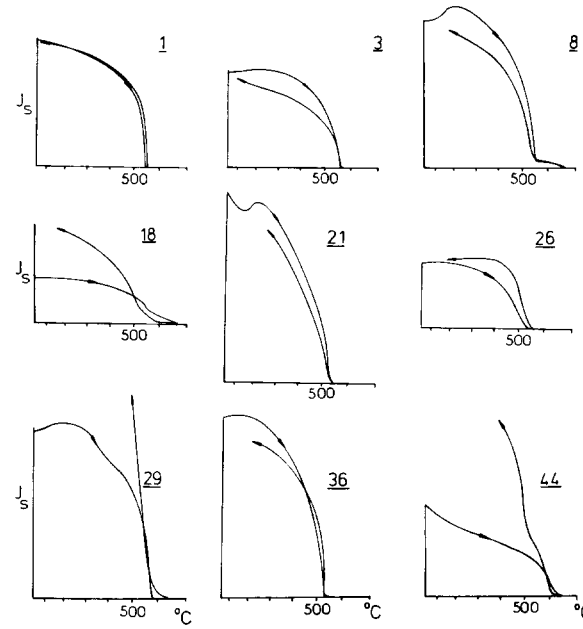


Fig.3. Thermomagnetic curves (saturation magnetisation J_s versus temperature) for nine Gardar lavas.

moment depends on factors other than the relative volume present.

During heat treatment samples typically undergo some change (Fig. 3, Table 1): a falling J_s may result from further production of hematite (of lower J_s [18]) from titanomagnetite, but the more common increase in J_s is more difficult to explain, and of several possible causes [19] only internal ordering of ferrimagnetic phases would appear to be applicable here.

6. Thermal demagnetisation studies

Thermal demagnetisation experiments were concentrated on those sites found from the petrologic examination to carry abundant hematite. Following large steps to 500°C, the demagnetisation interval was reduced to 30°C intervals to 590°C and 20°C intervals to 670°C to examine in detail any magnetic changes above the Curie point of any titanomagnetite present, and close to the Curie point of hematite. Behaviour of typical pilot samples is illustrated in Fig. 4.

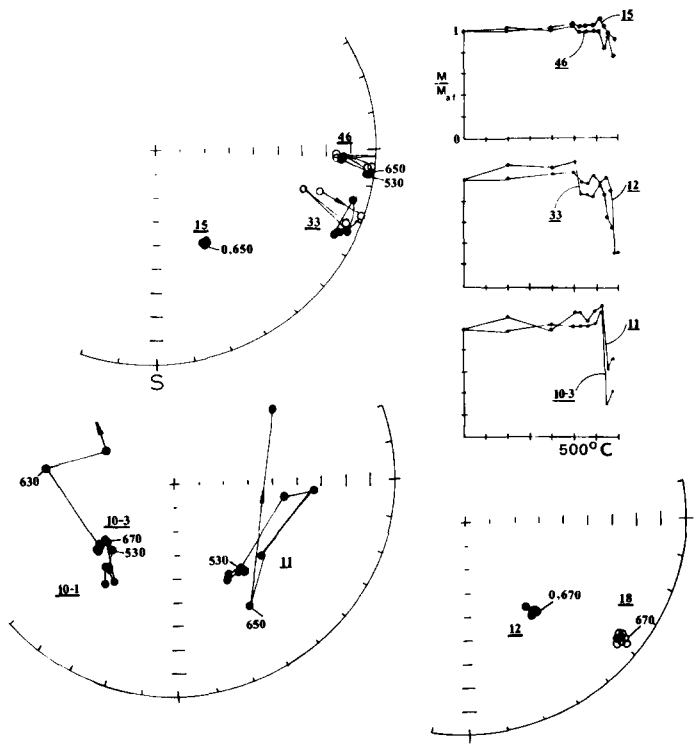


Fig. 4. Behaviour of directions of magnetisation and magnetic moments of some pilot samples with progressive thermal demagnetisation. The samples have already been cleaned in a.f. fields between 800 and 1400 Oe. Temperatures in °C are indicated and site numbers are underlined.

In the majority of pilot samples no significant change in direction was observed during progressive treatment, and directions measured at 670°C were identical to those at room temperature; in these samples there is therefore no evidence for two components with different ranges of blocking temperatures. A few samples (sites 9, 14) proved largely unstable to thermal demagnetisation, and several pilots exhibited small systematic movements with progressive treatment. All samples from these sites were accordingly demagnetised at a temperature selected from treatment of the pilot. The site means are included in Table 1 where they differ significantly at the 95% confidence level from site means computed from a.f. data (sites 10, 13, 14, and 22). No samples from the lower or upper groups gave directions after thermal treatment which were significantly different from those derived from prior a.f. treatment.

7. Magnetic stratigraphy and palaeomagnetic pole positions

The a.f. and thermally cleaned directions of magnetisation yield a sequence of virtual geomagnetic poles; these are interpreted as recording the sequence of reversals of the geomagnetic field. Variations in magnetic parameters with altitude in the lava pile are illustrated in Fig. 5. The sequence of poles (Fig. 6) is a record of ancient secular variation. In the lower group there is a single reversal of magnetisation with two sites (4 and 9) with steep positive magnetic inclinations regarded as intermediate directions (Fig. 7) and suggesting that a reversal may have been missed in the period taken for the intervening sandstones to be deposited. These sites are excluded from the group mean yielding a palaeomagnetic pole at 239°E, 7°N (7 sites, $A_{95} = 10.8°$). In the upper group intermedi-

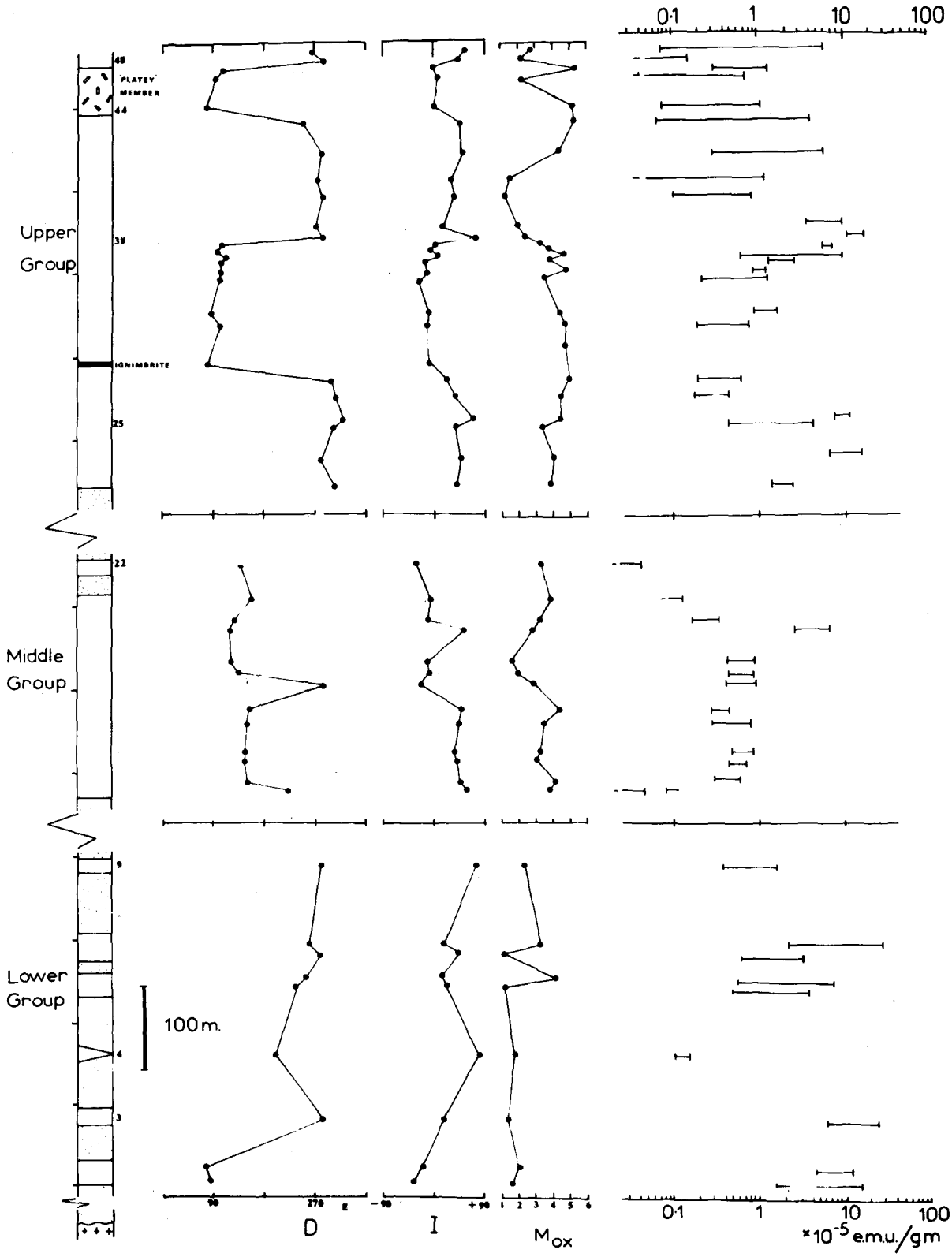


Fig. 5. Variation in magnetic declination, inclination, deuteric oxidation index (M_{ox}) and intensity of total NRM (assumed to have a logarithmic distribution) of Gardar lavas with position in the succession.

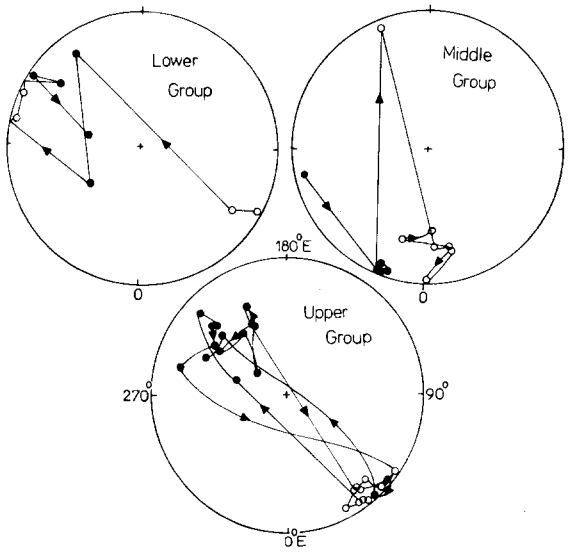


Fig. 6. Sequence of virtual geomagnetic poles through the Gardar lava succession. Stereographic projection, centred at N and S poles, open symbols are northern and closed symbols are southern hemisphere plots.

ate sites 26 and 38 also with steep positive magnetic inclinations are excluded from the group mean direction which yields a pole at 222°E, 16°N (24 sites, $A_{95} = 6.9°$). Of these four intermediate sites only 26 is comparable to the present geomagnetic field at the sampling locality (Fig. 7). Site 38 occurs between sites of opposite polarity (Table 1) suggesting that it indeed represents a transitional geomagnetic field direction. In the lower group where the effects of hematisation are least, the total natural remanent magnetisation (NRM) intensities of the two intermediate sites are also approximately an order lower than the remaining sites (Table 2); comparable low intensities are associated with young geomagnetic reversals (e.g. [20–22]).

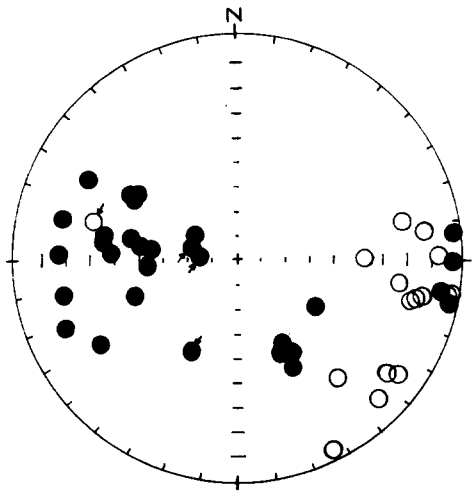


Fig. 7. Site mean directions of magnetisation from this study; directions excluded from site means of Table 1 are arrowed.

Directions of magnetisation from the middle group are more difficult to interpret because they do not agree with the other groups, and neither are the pole positions in accord with presently available North American results of comparable age. In these sites there is the most powerful case for low-temperature remagnetisation at later times because hematite content exceeds 10% of the opaques in all but two sites (Table 2) and is often very much greater. Against this is the apparent dominance of titanomagnetite-held remanence at site 19 and in some samples at other sites, and the comparability of magnetic properties with sites of the upper and lower groups where both polarities and intermediate directions are recorded. Also the sequence of virtual geomagnetic poles implies that lava units 11–15 (the second to the sixth lava in Fig. 6) were erupted within a short period of time to acquire an essentially instantaneous record of the geomagnetic field (pole A in Table 1), while lavas 17–22 were erupted at longer intervals, and site 16 is approximately antiparallel to the remainder of the group (17, 18, 20–22, palaeomagnetic pole B in Table 1).

The magnetic response of the (post-deuteric) hydrothermal activity can be resolved into components due to physical and chemical effects. The physical effect can be assessed using the analysis of Pullaiah et al. [38]. Since the thermal demagnetisation results all show that these lavas possess appreciable components with blocking temperature above 500°C, it is unlikely that hydrothermal temperatures persisted for long enough to cause remagnetisation. Indeed for blocking temperatures of 550°C a temperature of 520°C for 1 m.y. would be necessary to unblock the remanence [38]; lower temperatures would need to be attained for very much longer and temperatures of 350°C for 100 m.y. could only unblock the remanence with blocking temperatures below 490°C. It is no doubt for this reason that the successions comprising

the upper and lower groups of lavas retain a complete record of several episodes of opposite polarity (Fig. 5).

The effect of chemical changes, observed especially in the middle group, is more difficult to assess, but the evidence from thermal demagnetisation shows that the bulk of the lavas have only a single high blocking temperature component. It is very unlikely that these could have been acquired below deuteric temperatures, and it is noteworthy that since no progressive direction movement is observed in these rocks, they are quite different from highly oxidised Old Red Sandstone lavas where a case can be made from thermal demagnetisation studies [39] for more than one component of magnetisation residing in the same sample. This would explain why slight differences in direction are sometimes observed between high- and low-coercivity samples, but large differences are never observed, and the site means evaluated by a.f. and thermal cleaning are identical or differ by only a few degrees.

It appears therefore, that the sequence of virtual geomagnetic poles from the middle group records either an apparent polar wander excursion, or a series of poles consequent upon a geomagnetic field reversal; the latter explanation is less likely because the higher units in this group are separated by sediments and their period of eruption appears to have spanned a considerable period of time. The group illustrates a major problem encountered in defining Precambrian apparent polar wander (a.p.w.) curves since from their position these poles might be correlated with North American palaeomagnetic poles from the metamorphic Grenville Province dated by regional cooling ages in the range 1100–900 m.y., but are quite unlike any other data assigned to the interval 1300–1470 m.y. (section 9).

8. Geomagnetic field behaviour

We cannot yet be certain of the normal or reverse sense of the palaeomagnetic poles given in Table 1 because the very high rates of apparent polar movement in Late Precambrian-Cambrian times [23] leave an ambiguity in the interpretation of older a.p.w. curves. However, there are gross similarities in the Precambrian a.p.w. curves for the African, Laurentian and Scandinavian shields [24] and if the interpretation by McElhinny et al. [23] of the Late Precambrian-Cambrian a.p.w. curve for Gondwanaland is correct, the westerly directed magnetisations are south-seeking or reversed and the easterly directed magnetisations are north-seeking or normal.

With a minimum of five complete reversals in the upper and lower groups and incomplete evidence suggesting five more, the palaeomagnetic survey defines an interval of high reversal frequency analogous to that in Tertiary and Recent times. The sequence of virtual geomagnetic poles also suggests a complexity in the geomagnetic field with an analogue in Tertiary times. Thus in the upper group the normal poles are not the precise antipoles of the reversed poles; there is a significant difference of 15–20° between the normal and reversed field axis (Fig. 6). This difference cannot be a short-term effect because it persists over at least four polarity intervals, and it is similar to differences noted between normal and reversely magnetised dykes [25] in the Scottish Tertiary igneous province (50–60 m.y. old). In essence it is a consequence of the westerly directed magnetisations having steeper inclinations than the easterly directed magnetisations (Table 1), and would result from a displacement of the reversed dipole source along the rotational axis with respect to the normal dipole source; we cannot test from these data alone whether the field source was axial.

The oxidation-polarity correlation [26–28] has been identified in a considerable number of Phanerozoic rock units, and although no explanation exists for the phenomenon, it remains of immediate importance because it implies that processes within the earth's core affect the chemistry of basaltic magmas fractionated from the uppermost part of the mantle. This Proterozoic lava succession has a high reversal frequency and the opaque petrologic data of Table 2 provide a suitable opportunity for the concept to be tested in Precambrian times. In Fig. 8 the distributions of magnetite oxidation number and occurrence of sulphides are documented as histograms for the sites with easterly and westerly directed magnetisations; there is no significant difference in ilmenite content between the two groups. Although there is a spread of oxidation states for both groups, there is a limited tendency for samples from the easterly group to fall in the higher oxidation groups, and one of the samples in this group with lowest oxidation state is in any case a subaqueous lava [11,17]; also less than 10% of

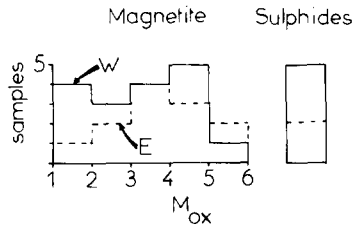


Fig. 8. Distribution of oxidation states and occurrence of sulphides in the Gardar lavas with easterly (E) and westerly (W) directed magnetisations.

the easterly group contain sulphides compared with nearly one-third of the westerly group; this statistical correlation provides a preliminary suggestion that oxidation-polarity correlations may be present in Proterozoic rocks. Finally, as a note of caution, it should be pointed out that the observed variability of magnetic properties within individual sites implies that within-lava variation of oxidation state may also be appreciable [29], and it is unlikely that one sample is representative of the lava from which it is derived.

9. Apparent polar wander movements relative to the Laurentian craton

With the results given here there are now 14 palaeomagnetic poles from igneous rock units within the Gardar igneous province (Table 3) of which the second (section 7) and possibly the fifth do not average components of secular variation and should be considered as a virtual geomagnetic poles only. Most of the remainder have ages closely defined by Rb–Sr isochrons and cross-cutting geological relationships so that they can be placed in sequence to define an a.p.w. path for the interval ca. 1400–1160 m.y. (Fig. 9).

Within this interval the palaeomagnetic data provide detailed information over three of the four relatively short intervals of time represented by magmatism within the Gardar Province [33]. The oldest event was probably represented by effusion of lavas and the subject of this paper, although the age relationship to the Grønnedal-Íka (1327 ± 17 m.y.) complex is not yet resolved. Slightly later alkaline bodies are Motzfedt (1310 ± 31 m.y.) and North Qôroq (1295 ± 61 m.y.) for which palaeomagnetic data do not yet exist. The next event, concentrated in the west of the province and contemporaneous with the Mackenzie igneous activity of the Canadian Shield [30], is represented by NNW-SSE dyke swarms and the Kûngnât complex (1245 ± 17 m.y.) from which poles 5–8 in Table 3 are derived. Finally at ca. 1175 m.y. intrusion of NE-SW-trending giant dykes commenced in the eastern part of the province to be followed by minor dykes and syenite complexes, and terminating at approximately 1160 m.y; these rocks are currently represented by poles 9–14 in Table 3.

North American palaeomagnetic results for the interval 1475–1160 m.y. are included in Table 4 together with the Greenland poles of Table 3 rotated

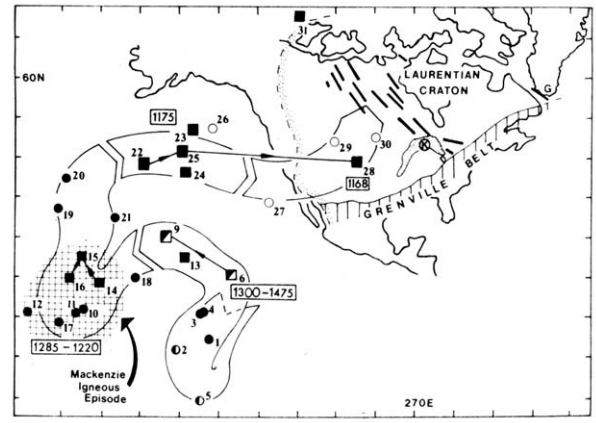


Fig. 9. Apparent polar wander (a.p.w.) path relative to the Laurentian craton for the interval 1475–1160 m.y. defined by poles from the Gardar Province of Greenland (squares) and North America (circles). Poles are listed with assigned ages in Table 3 and 4. Preferred ages (in m.y.) assigned to the a.p.w. path are indicated by the boxed numbers and a possible further complexity in the a.p.w. path is suggested by the middle group of lavas of this study, although the evidence for this is not considered sufficiently good for inclusion here. The Greenland poles are plotted with 95% confidence ovals in Piper [31]. The Mackenzie igneous episode in the Laurentian craton is defined by widespread intrusion of dolerite dykes indicated diagrammatically by bars. K is the outcrop of Keweenawan sediments (ca. 1160–1040 m.y.) and G is the location of the Gardar Province. The Laurentian craton is taken here to include the Precambrian regions of North America, Greenland and northwest Scotland. The Grenville mobile belt developed immediately after the apparent polar movements described here and underwent protracted uplift and cooling over a period of 200–300 m.y. The numbers refer to the poles as listed in Table 4, and lines of latitude and longitude are given at 10° intervals.

TABLE 3

Summary of palaeomagnetic results from the Gardar igneous province, South Greenland

Rock unit	Site/ samples	Palaeomagnetic pole		dΨ/dχ	Geologic age	Radiometric age (m.y.)	Reference
1 Gardar lavas, lower group	7/34	239.0°E	7.1°N	7.6/14.1	–	1600–1310	this paper
2 Gardar lavas, middle group (A)	5/24	339.7°E	1.8°N	3.6/ 5.5	<unit 1	1600–1310	this paper
3 Gardar lavas, middle group (B)	5/20	5.3°E	−26.3°N	7.2/14.2	<unit 1	1600–1310	this paper
4 Gardar lavas, upper group	24/113	222.1°E	15.6°N	4.8/ 9.0	<unit 3	1600–1310	this paper
5 BDO dolerite dykes	2/14	227.1°E	10.5°N	7.6/14.5	>unit 9	1243 ± 18 *, 1238 ± 18 *	[30]
6 NW lamprophyre dykes	9/51	206.4°E	3.2°N	5.1/10.1		<1330, >1245 1278–1254 **	[30]
7 NWW dolerite dykes	24/138	201.7°E	8.7°N	4.7/ 9.4	<unit 6	<1327, >1245	[30]
8 Gabbro ring dyke, Kûngnât	4/18	198.7°E	3.4°N	2.3/ 4.4	<unit 7	1245 ± 17 * 1233 ± 16 *	[30]
9 Hviddal Giant Dyke, Tugtutôq	7/38	215.3°E	33.4°N	7.6/12.2	<unit 5	1175 ± 9 *	[31]
10 Gabbro giant dykes, Tugtutôq	13/73	226.1°E	42.3°N	8.0/11.1	<unit 9	<1175, >1168	[31]
11 Narssaq, gabbro and ultramafics	4/22	225.4°E	31.6°N	7.8/12.1	<unit 9	<1175, >1168	[31]
12 NE-trending dyke swarms	47/264	223.9°E	36.4°N	4.1/ 6.1	<unit 11	>1168 ± 37	[30]
13 Ilímaussaq, marginal syenites	5/31	267.8°E	38.2°N	16.2/18.9	<unit 12	–	[32]
14 Ilímaussaq, fractionated rocks	6/32	238.6°E	72.7°N	37.1/41.0	<unit 13	1168 ± 21 * 1150 ± 28 *	[32]

Rb–Sr isochron (*) or whole-rock (**) ages on same or directly related rock unit to palaeomagnetic sample. The geochronological information is summarised in Blaxland et al. [9] and Patchett and Upton [41].

according to the 500-fathom (1000 m) continental reconstruction of Bullard et al. [36]. To derive a magnetic stratigraphy the poles have been listed in sequence to yield a simple a.p.w. path (Fig. 9); this involves a slightly subjective assessment of some of the age data, but in all cases, as for example between the apparently older Logan diabase (poles 20, 21, 1160 m.y.) and the apparently younger Hviddal Giant Dyke (pole 22, 1175 m.y.), the disagreements are within the errors attached to the age determinations.

The Gardar lavas (poles 6 and 9 in Table 4 and Fig. 9) give palaeomagnetic poles which fall 20–30° to the east of Mackenzie-related poles in common

TABLE 4

Palaeomagnetic poles for the Laurentian craton and magnetostratigraphy of the interval 1475–1160 m.y.

	Radiometric age (m.y.)	Palaeomagnetic study	Palaeomagnetic pole		Polarity	Listing
1	1475	Croker Island Complex	217.2°E	−5.4°N	W	1129
2	1410	Sherman Granite	209.0°E	−8.0°N	M	1172
3	1450–1400	Michikamau Anorthosite	215.3°E 218.0°E	0.6°N −1.5°N	W	1205 [42]
4	1375	St. Francis rocks	216.1°E	1.3°N	W	1375
5	1370	Sibley Group	216.0°E	−21.0°N	M	1470
6		Gardar lavas, lower group	222.8°E	10.6°N	M	Table 3
7		Gardar lavas, middle group (A)	322.9°E	1.7°N	–	Table 3
8		Gardar lavas, middle group (B)	348.3°E	−32.3°N	–	Table 3
9		Gardar lavas, upper group	206.3°E	20.4°N	M	Table 3
10	1250	Muskox intrusion	185.0°E	4.0°N	W	1115
11	1284	Coppermine Group	183.0°E	1.0°N	W	1472
12	1240	Mackenzie Diabase	171.0°E	1.0°N	W	1338
13	1245	BDO dykes	211.0°E	14.9°N	W	Table 3
14	1278–1254	Lamprophyre dykes	189.0°E	8.8°N	W	Table 3
15		NNW dolerite dykes	185.0°E	14.4°N	W	Table 3
16	1245, 1233	Kûngnât ring dyke	181.9°E	9.2°N	W	Table 3
17	1225	Sudbury dykes	179.0°E	−2.0°N	W	1120
18	1220	South Trap Range lavas	199.6°E	9.8°N	W	1374
19	1150	Gila County diabase sills	179.0°E	27.0°N	W	1320
20	1160	Logan diabase (Canada)	181.0°E	35.0°N	W	1268
21		Logan diabase (U.S.A.)	194.0°E	25.5°N	W	1185
22	1175	Hviddal Giant Dyke	200.6°E	38.3°N	W	Table 3
23		Gabbro giant dykes	213.1°E	46.7°N	W	Table 3
24		Narssaq Gabbro	211.1°E	36.1°N	W	Table 3
25		NE dykes, Tugtutôq	210.0°E	41.0°N	W	Table 3
26	1160	Logan diabase (Canada)	217.5°E	48.4°N	E	1309
27		South Trap Range lavas	232.5°E	28.0°N	E	1150
28		Ilímaussaq syenites	255.6°E	38.7°N	W	Table 3
29		Alona Bay lavas	250.0°E	44.0°N	E	1310
30		Baraga County dykes	260.5°E	45.0°N	E	[34]
31	1168, 1150	Ilímaussaq fractionated rocks	240.4°E	75.5°N	W	Table 3

Seal Group Redbeds (ca. 1300 m.y.) are not included in this compilation because of uncertainty as to their precise age of magnetisation. Muskox intrusion is listed before Coppermine Group because it is unconformably overlain by the latter although the assigned ages are not consistent with this. Pole 30 is derived from measured steep negative inclinations on material of probable Keweenawan age but stability was not tested by cleaning. Since the normal or reverse sense of the polarity given is not known, results are categorised by the directions of magnetisation being westerly directed (W) and yielding poles plotted in Fig. 7 or easterly directed (E) or mixed (M). The listing number is that given in the Ottawa Catalogue of Palaeomagnetic Directions and Poles [35], and all Greenlandic poles are corrected for Phanerozoic drift [36].

with 1300–1475 m.y. North American poles, with which there is noteworthy agreement. The polar shift from 6 to 9 during the period of eruption of the Gardar lavas may reflect the apparent polar movement towards the Mackenzie position and dated from North American results post-1370 m.y. and pre-1285 m.y. The results from igneous rocks related to the Mackenzie igneous episode define a quasi-static interval of constant polarity with extreme limits defined by the present data of 1285 and 1220 m.y. Following this episode the motion of the Laurentian Shield changed direction with a rapid movement towards the pole (Fig. 9). The youngest Gardar results (13 and 14, Table 3) correlate closely with several palaeomagnetic poles from the Keweenawan rocks of North America including the Alona Bay lavas from the lower part of the group([37], Table 4), and it is probable that the youngest Gardar results are equivalent in time only to the oldest Keweenawan lavas and sediments [32]. This dates the commencement of deposition at 1160 m.y. or marginally earlier than the 1140-m.y. estimate of Silver and Green [40]. Extensive results from Middle Keweenawan rocks define [37] a return of the Laurentian Shield to low latitudes by 1040 m.y.

The low palaeolatitude results of Table 4 will be sensitive to any inconsistencies in the continental reconstruction of Greenland and North America, and within the limits of errors assigned to the palaeomagnetic poles and differences in age, there are no systematic differences between the path defined by Greenland and North American data (Fig. 9). On these grounds we can conclude that the reconstruction of Bullard et al. [36] is applicable to Upper Proterozoic times within the limitations of the present data.

Acknowledgements

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